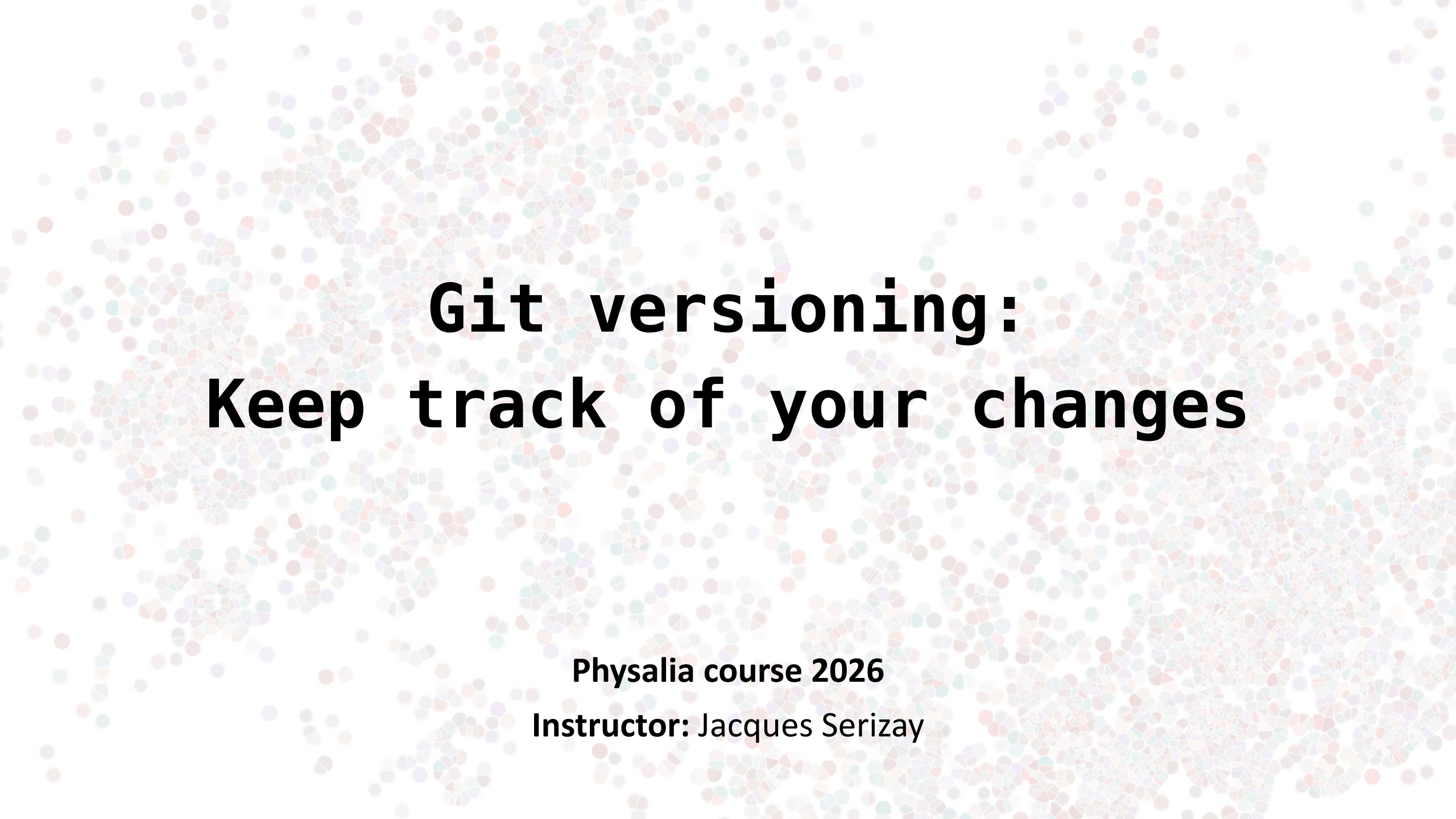


The background of the slide is a dense, abstract pattern of small, overlapping circles and triangles in various colors including red, blue, green, yellow, and purple. The pattern is more concentrated in the center and fades out towards the edges.

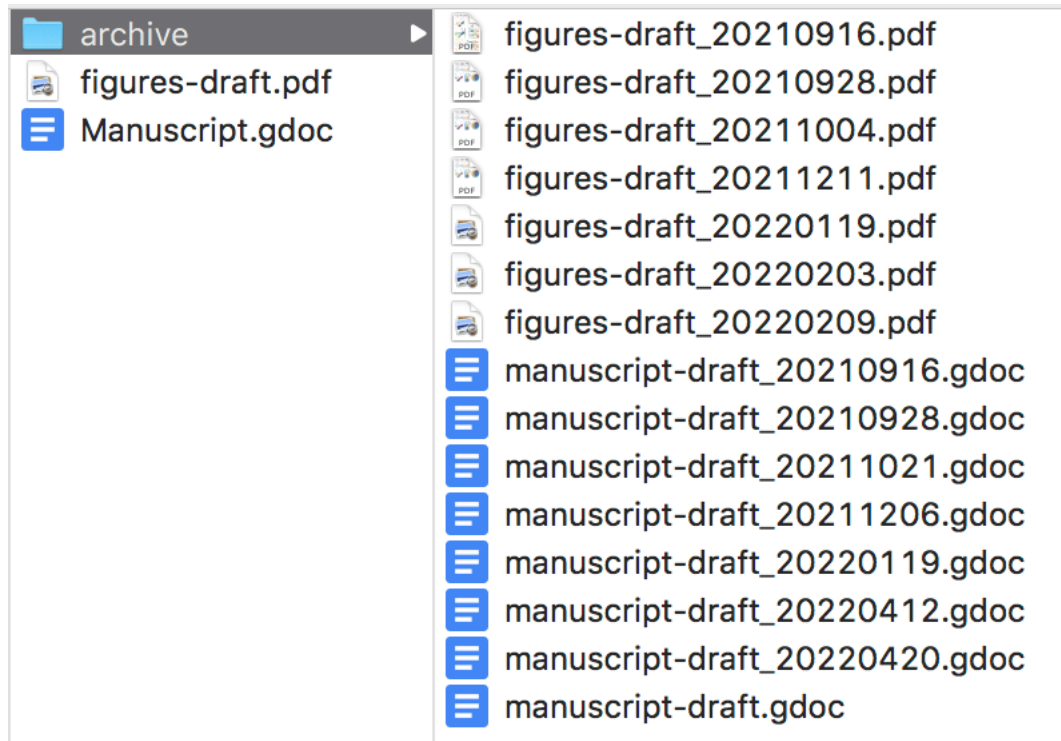
Git versioning: Keep track of your changes

Physalia course 2026

Instructor: Jacques Serizay

Keeping track of changes with git

- Ever time-stamped a file? Git does the same, for one or several files, and allows you to revert back to previous files or compare changes (for text files).



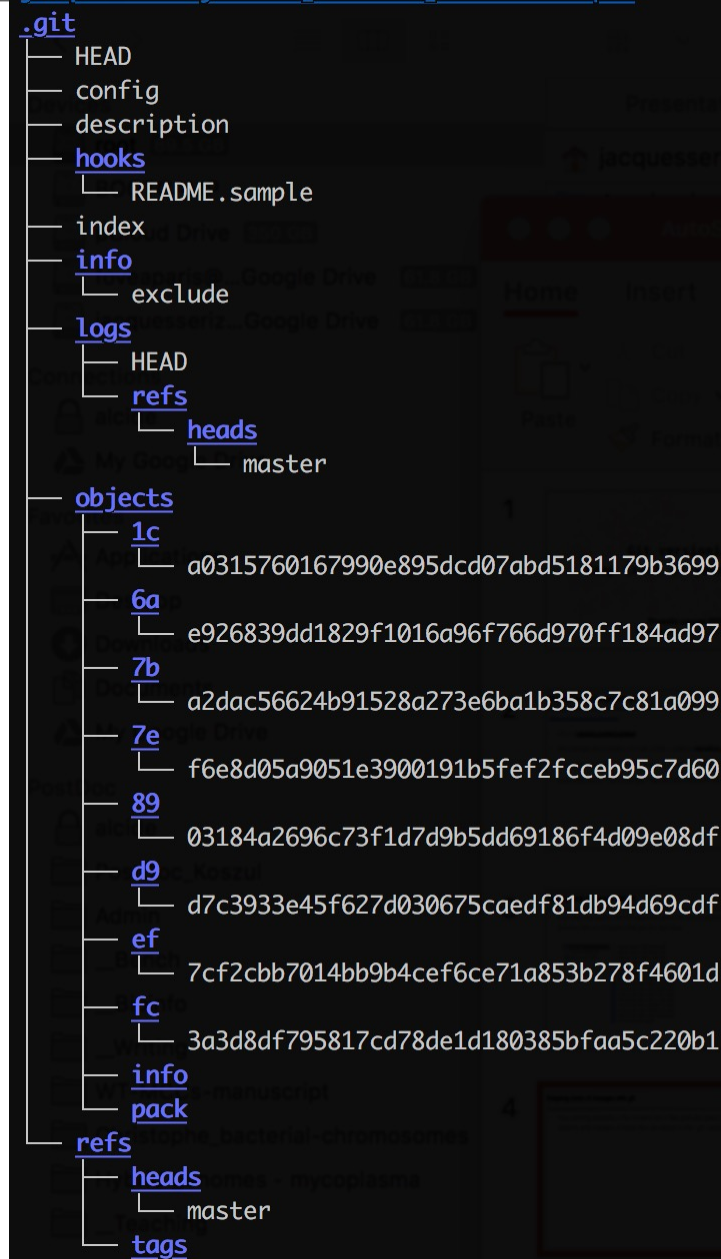
Keeping track of changes with git

<https://happygitwithr.com/>

- Git is a version control system.
- Git manages the evolution of a set of files – called a **repository** – in a sane, highly structured way.
- Its original purpose was to help groups of developers work collaboratively on big software projects.
- Essentially, this is “Track Changes” features from Microsoft Word on steroids.

Keeping track of changes with git

- Your working directory only contains the 2 files (pdf and gdoc) you are working on, and changes of these files are stored in the `.git` subfolder, in a compressed way.



Keeping track of changes with git

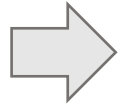
Git's main downside is... git is not user-friendly!

- It's painful to visualize changes, to browse history, to compare files, ...
- No native graphical interface

Git vs github

- Git is a software, locally installed on one's computer

```
$> git --help
```

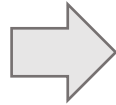


“Track Changes” features

Git vs github

- Git is a software, locally installed on one's computer

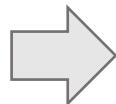
```
$> git
```



"Track Changes" features

- Github is an Internet hosting service, storing your local `git` repository (files AND changes) on-line ("remotely"):

```
https://github.com/
```



Dropbox features

Git vs github

- You are working on your manuscript draft in Word: you may rewrite sentences, resolve previous comments, reply to others still open, ... As long as you are working on your own, this works.
- Now you want to share it with others → you decide to put the document online as a Google Doc, allow access to others, and enable synchronization with your local Google Drive.
- Your colleagues will eventually rewrite/resolve/reply together with you (sometimes with conflicting changes!), and everyone will see who is doing what, etc.

The Google Doc is your “remote repository” and your local Google Drive folder is your “local repository”

Typically, both are continuously synced, whereas Git and Github are not sync-ed “live”, you do it manually and whenever you want.

Git vs github

Github, gitlab, bitbucket: all different internet hosting services doing the same thing:

1. Storing your local git repositories online, to make available to others.
2. Allow specific online-based features: issues and pull requests
3. Easy "time-travel" (i.e. browsing files back in time) with permanent links
4. Search function!

Git vs github

Github, gitlab, bitbucket: all different internet hosting services doing the same thing:

1. Storing your local git repositories online, to make available to others.

The screenshot shows the GitHub interface for the repository `js2264 / HiContacts`. The repository is public and has 1 fork and 0 stars. The `Code` button is highlighted, and a dropdown menu is open showing the following options:

- Clone**: A dropdown menu with three options: `HTTPS`, `SSH` (selected), and `GitHub CLI`. Below these options is the text "Use a password-protected SSH key."
- Open with GitHub Desktop**: A button with a GitHub Desktop icon.
- Download ZIP**: A button with a ZIP icon.

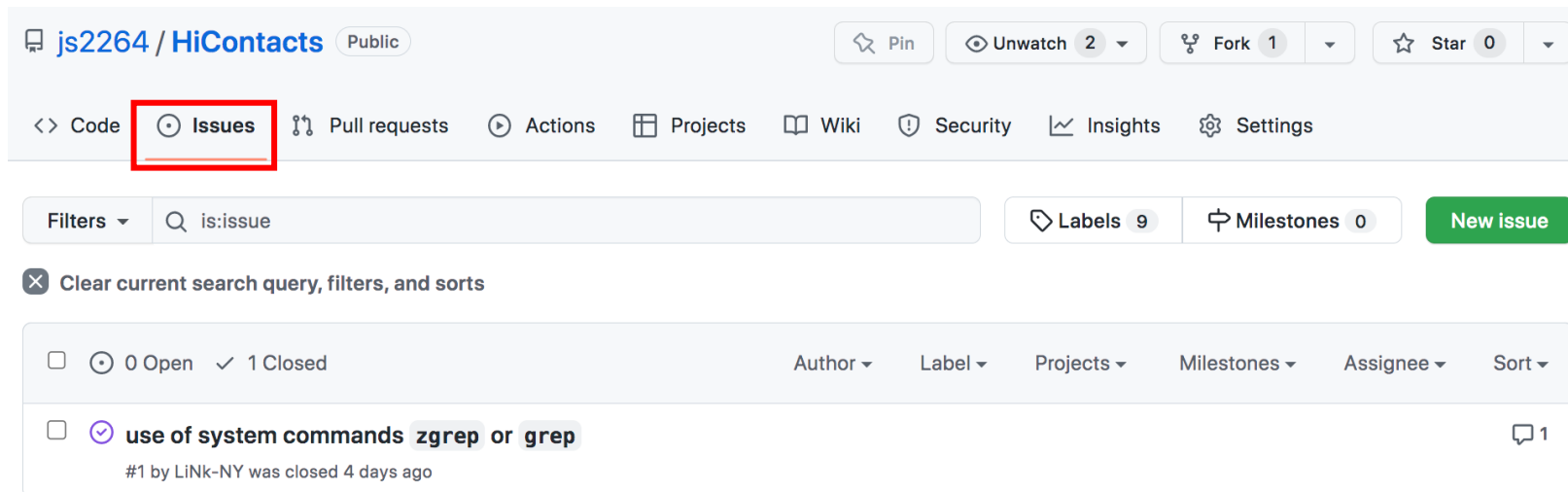
The repository details on the right show the following information:

- About**: HiContacts: R interface to cool files and analysis of Hi-C contacts. Link: js2264.com/HiContacts/.
- Readme**: A link to the README file.
- Unknown, MIT licenses found**: A link to the license information.
- 0 stars**: The number of stars the repository has.
- 2 watching**: The number of users watching the repository.
- 1 fork**: The number of forks the repository has.
- Releases**: No releases published. Link: [Create a new release](#).
- Packages**: No packages published. Link: [Publish your first package](#).

Git vs github

Github, gitlab, bitbucket: all different internet hosting services doing the same thing:

2. Allow specific online-based features: issues and pull requests



Git vs github

Github, gitlab, bitbucket: all different internet hosting services doing the same thing:

2. Allow specific online-based features: issues and pull requests

The screenshot shows the GitHub interface for the repository 'koszullab / hicstuff'. The 'Pull requests' tab is highlighted with a red box. The repository is public and has 8 forks and 28 stars. The 'Pull requests' tab shows a list of pull requests with the following details:

Checkbox	Icon	Title	Status	Author	Label	Projects	Milestones	Reviews	Assignee	Sort
☐		0 Open	✓ 11 Closed							
☐		add prefix to fig_dir	#60 by js2264 was merged on Jan 27 • Approved							1
☐		Replace travis with github workflow	#59 by cmdoret was merged on Jan 21							1
☐		Digest map	#46 by ABignaud was merged on Apr 19, 2021 • Approved							
☐		Move to BSD 3-Clause License	#44 by cmdoret was merged on Oct 15, 2020							
☐		Throw instead of returning input in normalization	#42 by scovit was merged on Apr 30, 2020							2

Git vs github

Github, gitlab, bitbucket: all different internet hosting services doing the same thing:

3. Easy “time-travel” (i.e. browsing files back in time) with permanent links

The screenshot displays the GitHub interface for the repository `koszullab / hicstuff`, which is public and forked from `baudrly/hicstuff`. The repository has 8 forks and 28 stars. The navigation bar includes links for Code, Issues (3), Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings.

The commit history for `hicstuff / README.md` is shown, with a vertical timeline on the left. The commits are as follows:

- Commits on Oct 10, 2022**
 - Add instructions for conda and docker installation** (Verified) by `cmdoret` committed 7 days ago. Commit hash: `89891c3`.
- Commits on Jan 20, 2022**
 - Replace travis status badge with gh-workflow** (Verified) by `cmdoret` committed on Jan 20. Commit hash: `5f606be`.
- Commits on Aug 24, 2021**
 - update travis badge in readme** by `cmdoret` committed on Aug 24, 2021. Commit hash: `ca735b7`.
- Commits on May 19, 2021**
 - fix pairs specification in readme** by `cmdoret` committed on May 19, 2021. Commit hash: `17b5a21`.

Git vs github

Github, gitlab, bitbucket: all different internet hosting services doing the same thing:

3. Easy “time-travel” (i.e. browsing files back in time) with permanent links

 **koszullab / hicstuff** Public

forked from baudrly/hicstuff

100644 390 Lines (287 sloc) 18.1 KB

Edit Pins

Watch 8

Fork 8

Star 28

Code

Issues 3

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

hicstuff / README.md

100644 390 Lines (287 sloc) 18.1 KB

Raw Normal view History

	Initial commit	7 years ago	1	# hicstuff
	Update README.md	7 years ago	2	
	Packaging, merging hicstuff and yahcp	4 years ago	3	[![PyPI version](https://badge.fury.io/py/hicstuff.svg)](https://pypi.org/project/hicstuff/)
			4	[![PyPI - Python Version](https://img.shields.io/pypi/pyversion/hicstuff.svg)](https://img.shields.io/pypi/pyversion/hicstuff.svg)
	Add instructions for conda and docker installation	7 days ago	5	[![install with bioconda](https://img.shields.io/badge/install%20with%20bioconda-bioconda.svg)](https://bioconda.github.io/recipes/hicstuff/README.html)
	Replace travis status badge with gh-workflow	9 months ago	6	[![Build Status](https://github.com/koszullab/hicstuff/actions/workflows/main.yml/badge.svg)](https://github.com/koszullab/hicstuff/actions/workflows/main.yml)
	Add zenodo DOI badge to README	3 years ago	7	[![DOI](https://zenodo.org/badge/DOI/10.5281/zenodo.2620601.svg)](https://doi.org/10.5281/zenodo.2620601)
	Coverage badge	4 years ago	8	[![codecov](https://codecov.io/gh/koszullab/hicstuff/branch/main/graph/badge.svg)](https://codecov.io/gh/koszullab/hicstuff/branch/main)
	Packaging, merging hicstuff and yahcp	4 years ago	9	[![Read the docs](https://readthedocs.org/projects/hicstuff/badge/?version=latest)](https://readthedocs.org/projects/hicstuff/)
	Binder link	4 years ago	10	[![Binder](https://mybinder.org/badge_logo.svg)](https://mybinder.org/v2/gh/koszullab/hicstuff/main?filepath=README.md)
	Packaging, merging hicstuff and yahcp	4 years ago	11	[![Code style: black](https://img.shields.io/badge/code%20style-black-000000.svg)](https://github.com/ambv/black)
	Update README.md	7 years ago	12	
	update readme with new pipeline commands	4 years ago	13	A lightweight library that generates and handles Hi-C contact matrices
			14	
			15	The goal is to make generation and manipulation of Hi-C matrices easy
	Packaging, merging hicstuff and yahcp	4 years ago	16	
	Fix hicstuff pipeline help in README, add TOC and list p...	4 years ago	17	## Table of contents
			18	
	Add instructions for conda and docker installation	7 days ago	19	* [Installation](#installation)
			20	* [pip](#pip)
			21	* [conda](#conda)
			22	* [docker](#docker)
			23	* [Usage](#usage)
			24	* [Full pipeline](#full-pipeline)
			25	* [Individual components](#individual-components)
			26	* [Library](#library)
			27	* [Connecting the modules](#connecting-the-modules)
			28	* [File formats](#file-formats)
			29	* [Contributing](#contributing)

Github and organizations

- A public repository is readable by the world. The owner can grant higher levels of permission to others, such as the ability to push commits.
- A private repository is invisible to the world. The owner can grant read, write (push), or admin access to others.
- There is also a notion of an organization, which can be useful for managing repository permissions for entire teams of people.

Github and organizations

- A public repository is readable by the world. The owner can grant higher levels of permission to others, such as the ability to push commits.
 - A private repository is invisible to the world. The owner can grant read, write (push), or admin access to others.
 - There is also a notion of an organization, which can be useful for managing repository permissions for entire teams of people.
- Github has been re-purposed by the data science community.
- We now also use it, IN COLLABORATION WITH A CLOUD-BASED GIT HOST, to manage the diverse collection of files that make up typical data analytical projects, which often consist of data, figures, reports, and, yes, source code.

Github and organizations



R infrastructure

Overview

Repositories 136

Projects

Packages

People 15

Pinned

 devtools

Public

Tools to make an R developer's life easier

R

2.2k

747

 roxygen2

Public

Generate R package documentation from inline R comments

R

521

208

 testthat

Public

An R 🍷 to make testing 🍷

R

810

299

 pkgdown

Public

Generate static html documentation for an R package

R

636

307

 usethis

Public

Set up commonly used 🍷 components

R

718

233

 httr

Public

httr: a friendly http package for R

R

959

2k



tidyverse

The tidyverse is a collection of R packages that share common principles and are designed to work together seamlessly

361 followers

http://tidyverse.org

Verified

Overview

Repositories 37

Projects

Packages

People 27

Pinned

 dplyr

Public

dplyr: A grammar of data manipulation

R

4.2k

1.5k

 ggplot2

Public

An Implementation of the Grammar of Graphics in R

R

5.6k

1.9k

 readr

Public

Read flat files (csv, tsv, fwf) into R

R

924

278

 purrr

Public

A functional programming toolkit for R

R

1.1k

229

 tidyr

Public

Tidy Messy Data

R

1.2k

393

 rvest

Public

Simple web scraping for R

R

1.4k

332



Bioconductor

Software for the analysis and comprehension of high-throughput genomic data

49 followers

https://bioconductor.org

Overview

Repositories 260

Projects 2

Packages

People 2

Pinned

 Contributions

Public

Contribute Packages to Bioconductor

115

32

 BIOCStickers

Public

Stickers for some Bioconductor packages - feel free to contribute and/or modify.

R

110

77

 bioconductor_docker

Public

Docker Containers for Bioconductor - NEW!

Shell

50

18

 bioconductor.org

Public

Source code for the Bioconductor website

HTML

16

33

 pkgrevdocs

Public

Training Material for Community Reviewers

TeX

6

11

 BBS

Public

The Bioconductor Build System

Python

7

6



rOpenSci

190 followers

Berkeley, CA

https://ropensci.org/

@rOpenSci

info@ropensci.org

Verified

Overview

Repositories 328

Projects 1

Packages

People 79

README.md

👋 Welcome to the rOpenSci Project on GitHub. Our repositories are currently distributed across a few different organizations.

You can explore our [resources page](#) to find what you're after, it includes a list of all our GitHub organizations.

Also refer to our [code of conduct](#).

Pinned

 software-review

Public

rOpenSci Software Peer Review.

R

249

103

 software-review-meta

Public

For organizing projects related to rOpenSci Software Peer Review

11

3

 dev_guide

Public

rOpenSci Packages: Development, Maintenance, and Peer Review

R

133

49

 rtweet

Public

R client for interacting with Twitter's [stream and REST] APIs

R

751

202

How to set things up

1. Register for a GitHub account
2. **MAKE SURE YOU ARE RUNNING R/RStudio/BiocManager WITH THE APPROPRIATE VERSION!**
3. Install Git: <https://git-scm.com/downloads>, or:
 - Linux: ``sudo apt-get install git``
 - Mac: ``xcode-select --install`` (needs Xcode)
 - Windows: ``choco install git.install`` (needs Chocolatey)

How to set things up

1. Register for a GitHub account
2. **MAKE SURE YOU ARE RUNNING R/RStudio/BiocManager WITH THE APPROPRIATE VERSION!**
3. Install Git: <https://git-scm.com/downloads>, or:
 - Linux: ``sudo apt-get install git``
 - Mac: ``xcode-select --install`` (needs Xcode)
 - Windows: ``choco install git.install`` (needs Chocolatey)

4. Configure **git** from R (easy)

```
usethis::git_sitrep()
gert::git_config_global_set("user.name", "Jacques")
gert::git_config_global_set("user.email", "jacquesserizay@gmail.com")
usethis::git_vaccinate()
usethis::use_git()
usethis::use_git_ignore("*.Rproj")
```

Don't forget to make an initial commit!

5. Configure github (harder):

- Git can communicate with a remote server using one of two protocols, HTTPS or SSH, and the different protocols use different credentials.
- We must use SSH, as this means of communications is required later on, for Bioconductor submission.
- For that, we need to generate a pair of private/public SSH keys, connect to GitHub from R and push the public key to GitHub.

How to set things up

5. Configure github (harder):

- Git can communicate with a remote server using one of two protocols, HTTPS or SSH, and the different protocols use different credentials.
- We must use SSH, as this means of communications is required later on, for Bioconductor submission.
- For that, we need to generate a pair of private/public SSH keys, connect to GitHub from R and push the public key to GitHub.

```
gh::gh_whoami()
usethis::gh_token_help()

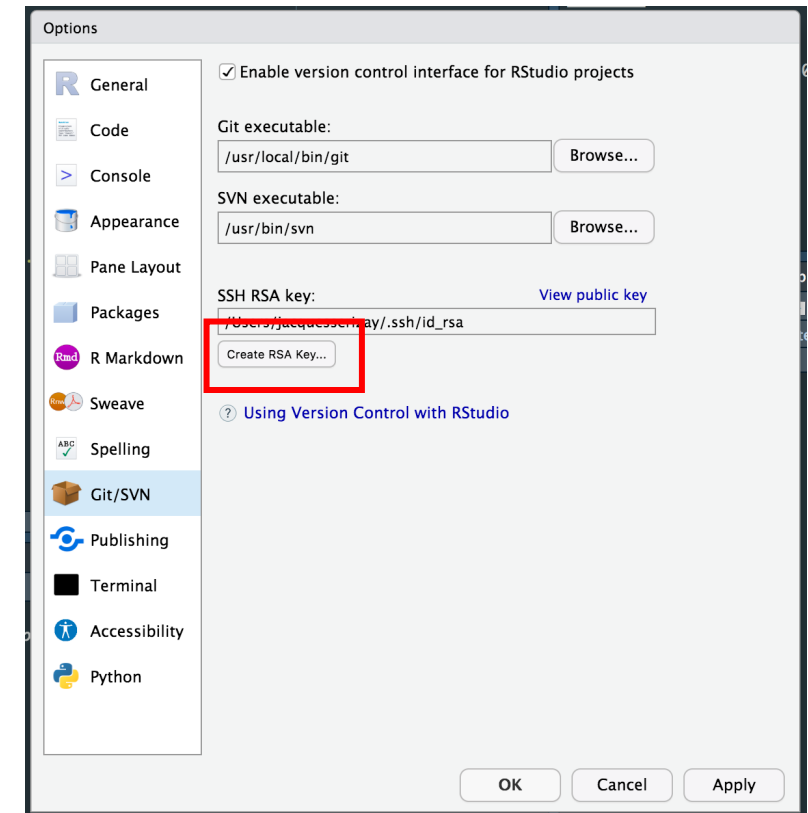
#Create a GH PAT
usethis ::create_github_token()
gitcreds ::gitcreds_set()
#Paste your GitHub token, not your password!

#Add an SSH key to GitHub
usethis ::use_github()
credentials::ssh_setup_github()
```

How to set things up

5. Configure github (harder):

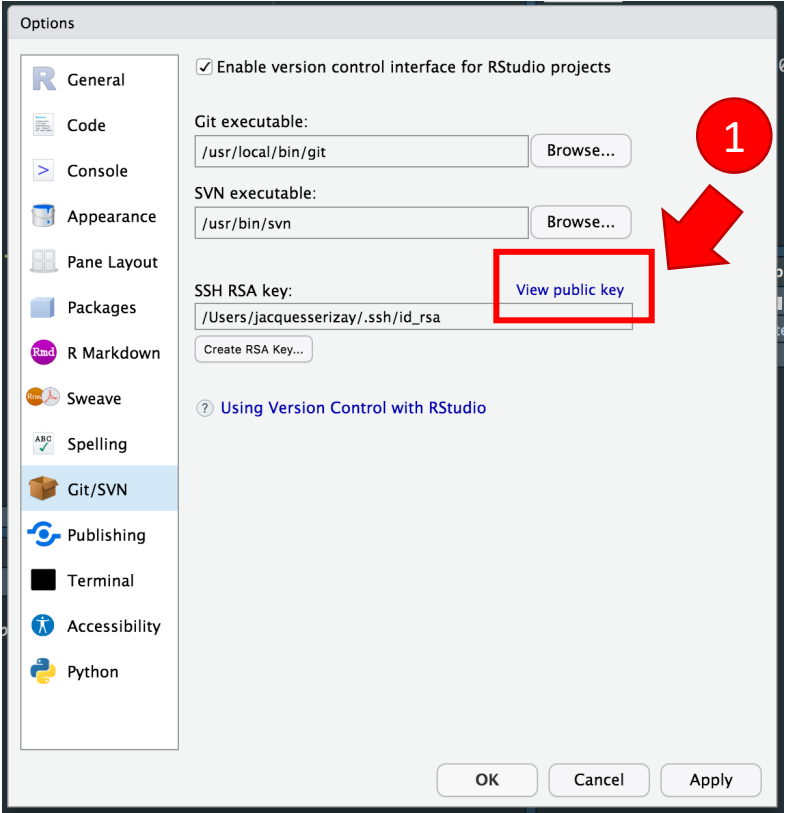
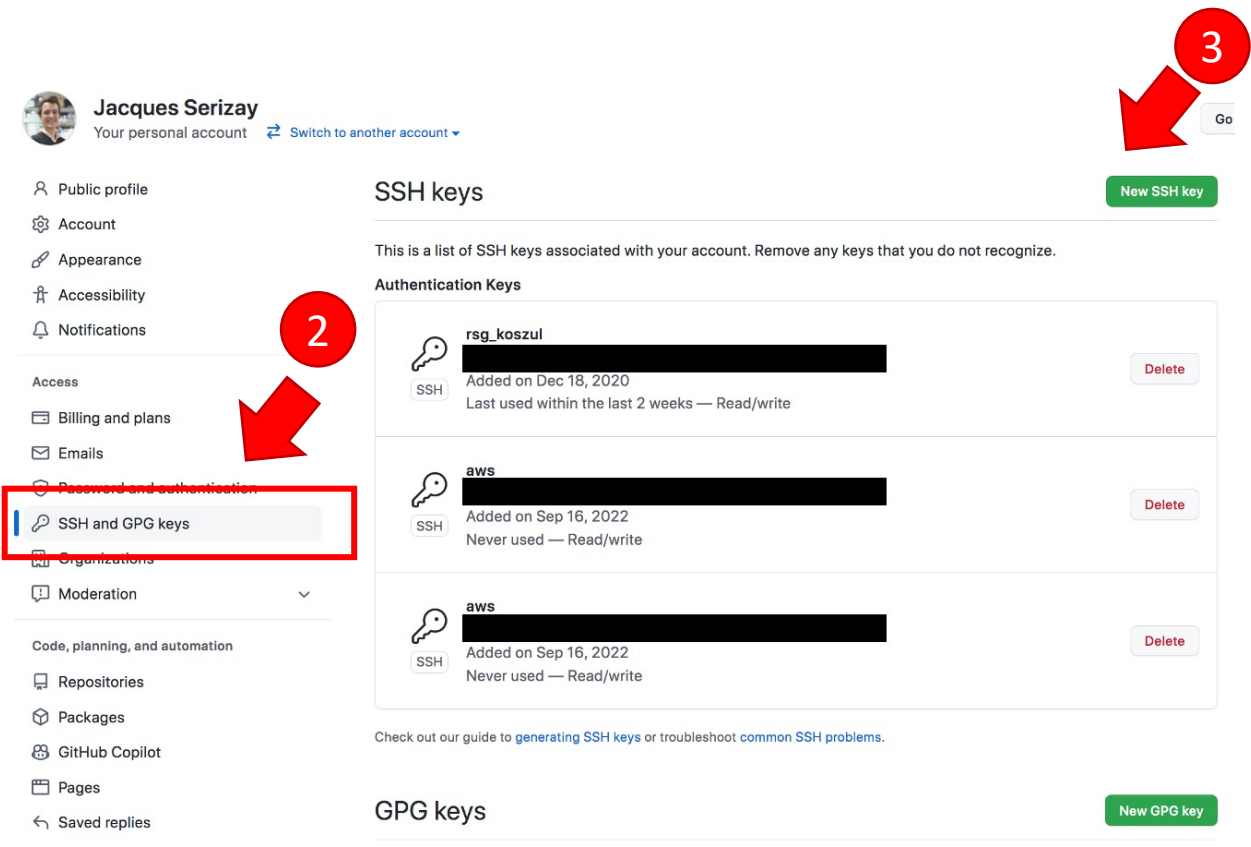
- Fallback approach: create a key in RStudio



How to set things up

5. Configure github (harder):

- Then, we need to upload the public key to GitHub



How to set things up

5. Configure github (harder):

- Test it with ``ssh -T git@github.com``

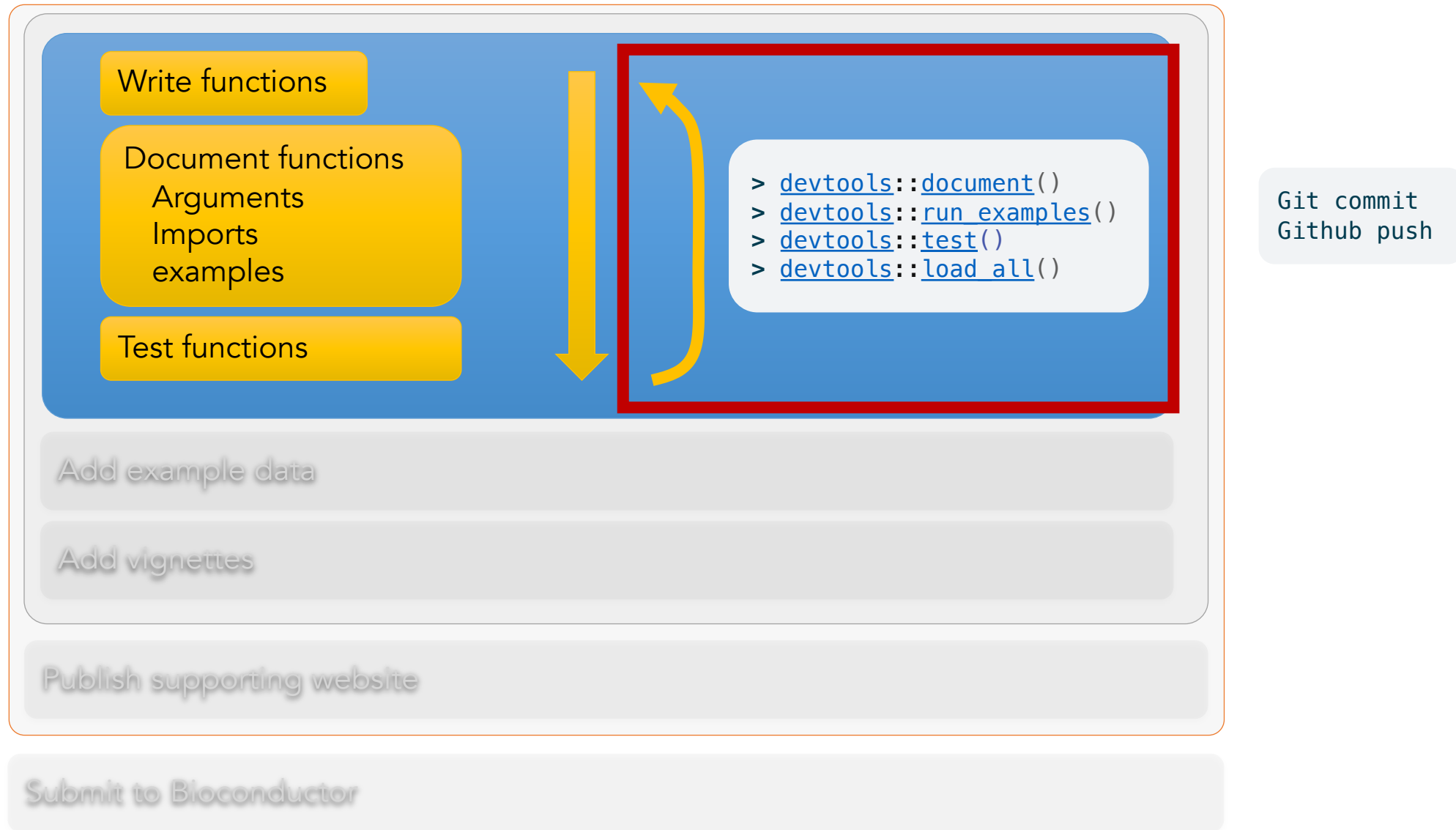
```
jacquesserizay@LOCAL[13:54:27]:~ $ ssh -T git@github.com  
▶Hi js2264! You've successfully authenticated, but GitHub  
does not provide shell access.
```


How to set things up

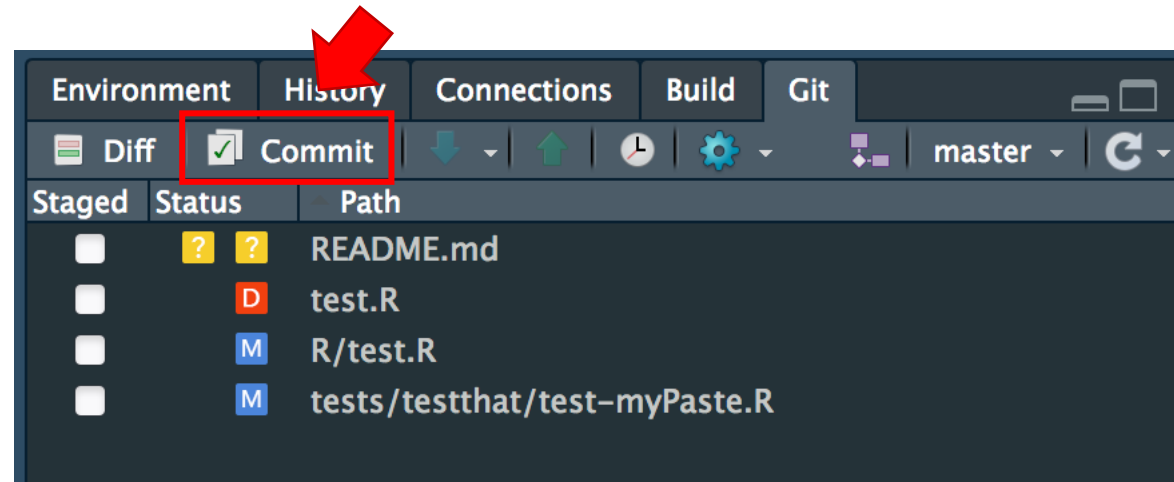
If you are still struggling...

<https://happygitwithr.com/usage-intro.html#usage-intro>

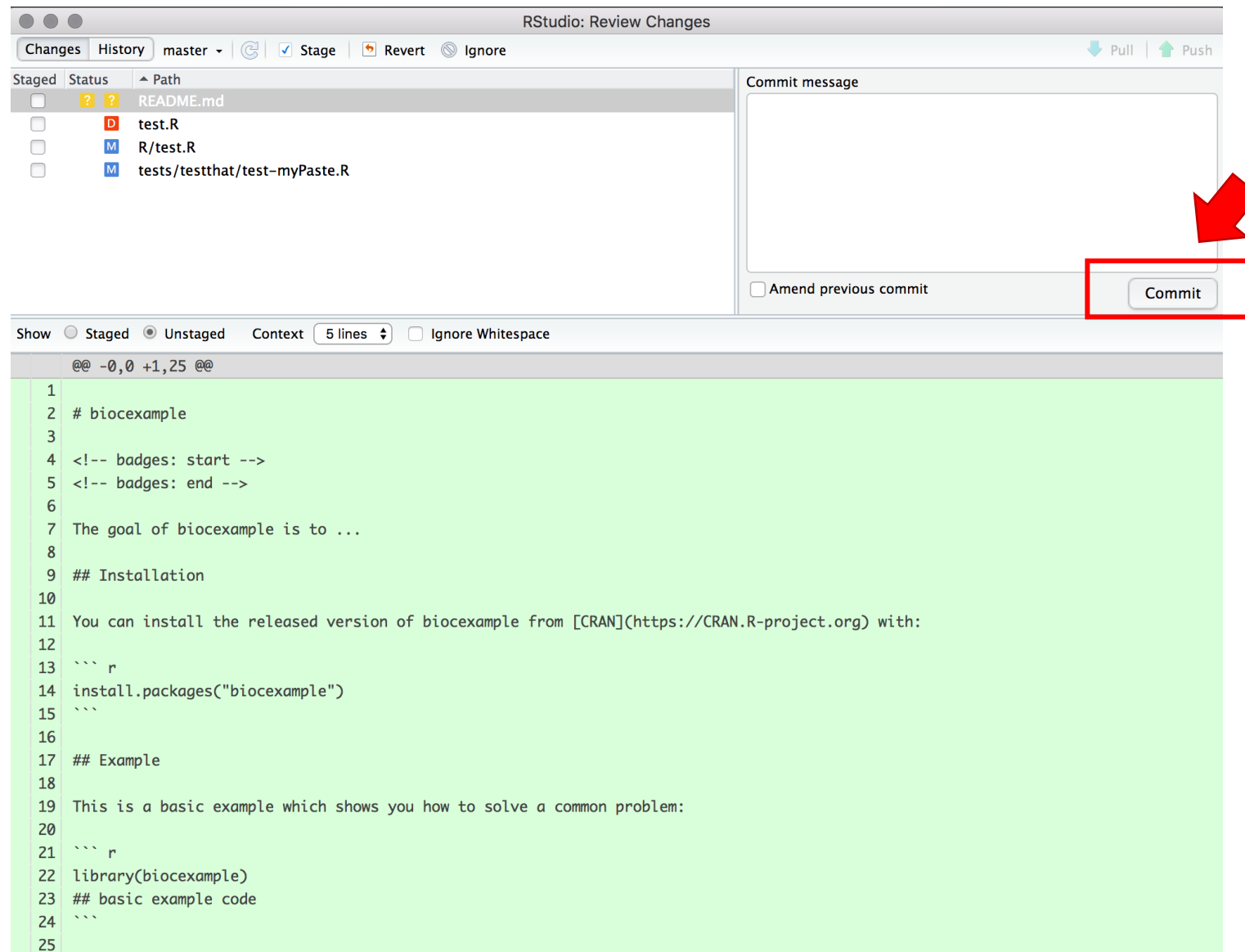
Lather-rinse-repeat



Lather-rinse-repeat



Lather-rinse-repeat



Lather-rinse-repeat

